

# Towards Benchmarking of Psychological Time Series

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FGME JuMi Retreat · October 2026

Björn Siepe

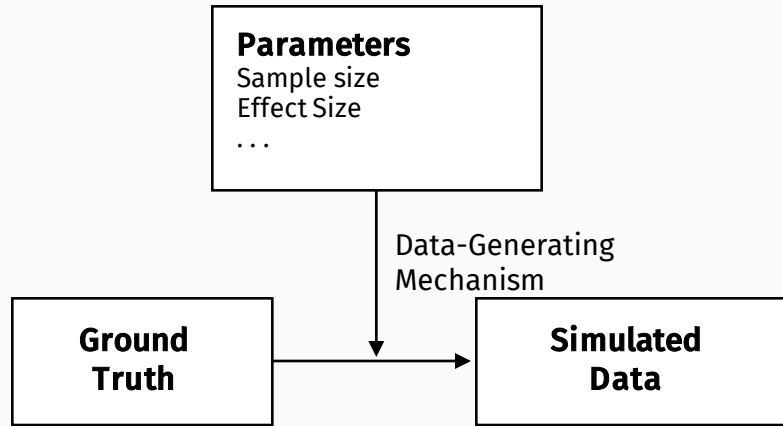
Psychological Methods Lab, Department of Psychology, Marburg University

# Simulation studies

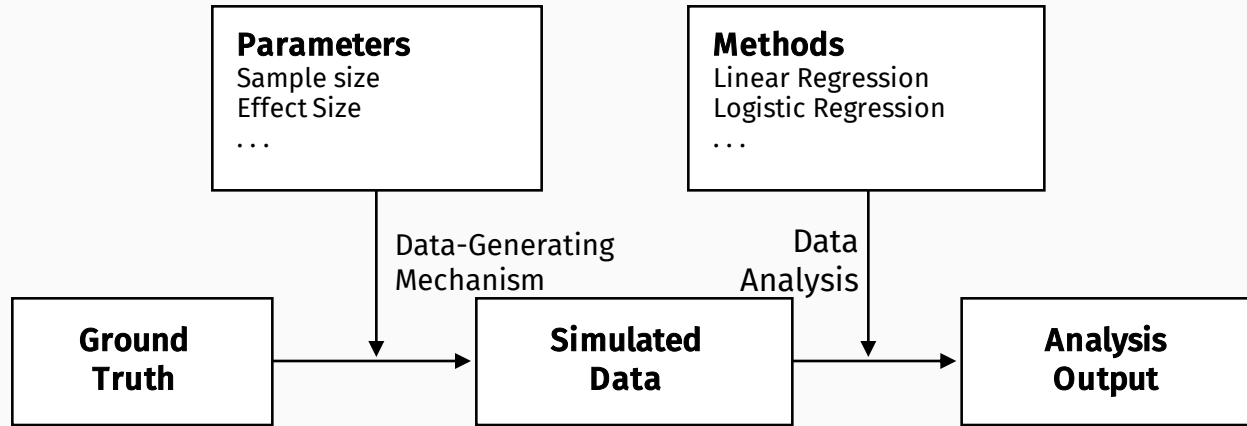
# Simulation studies

**Ground  
Truth**

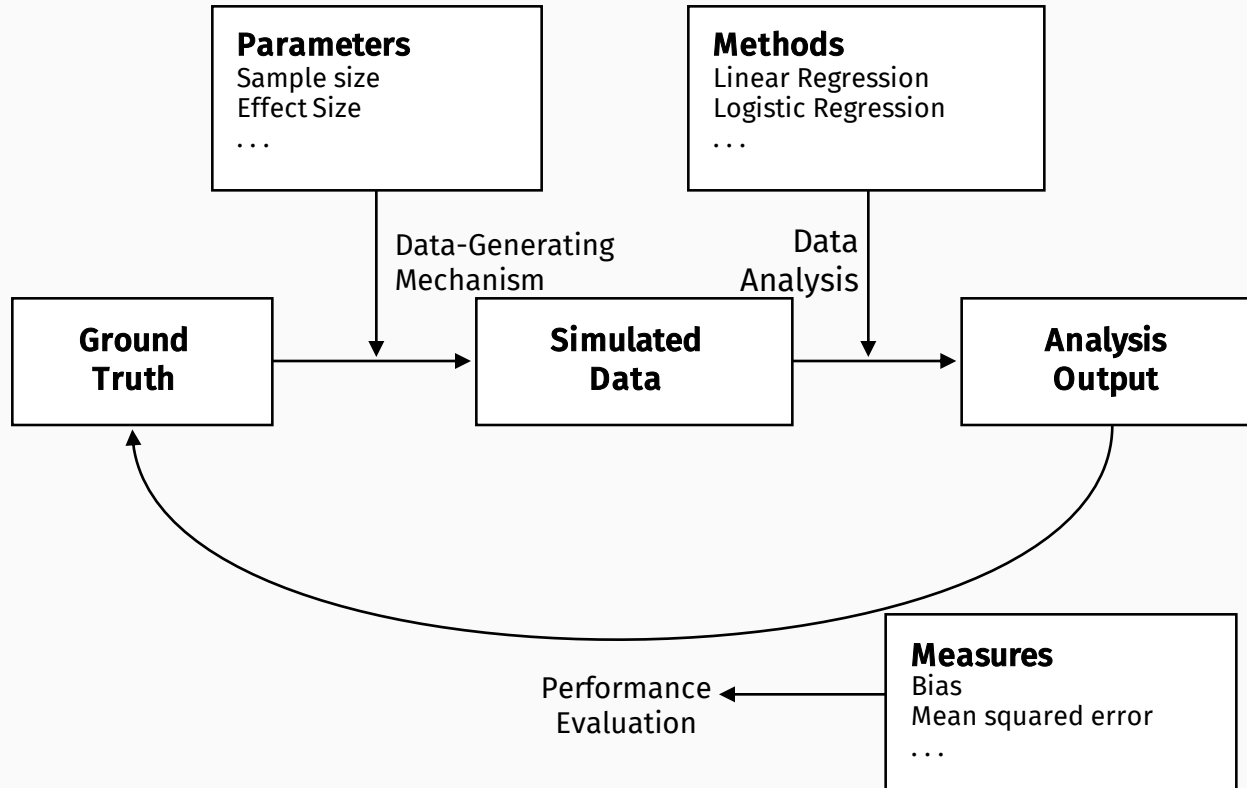
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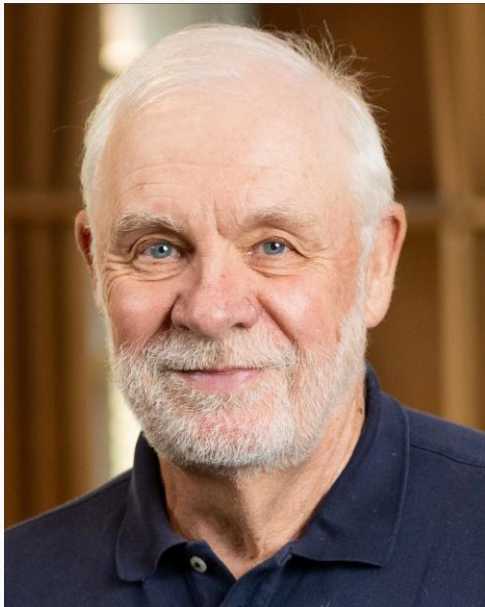


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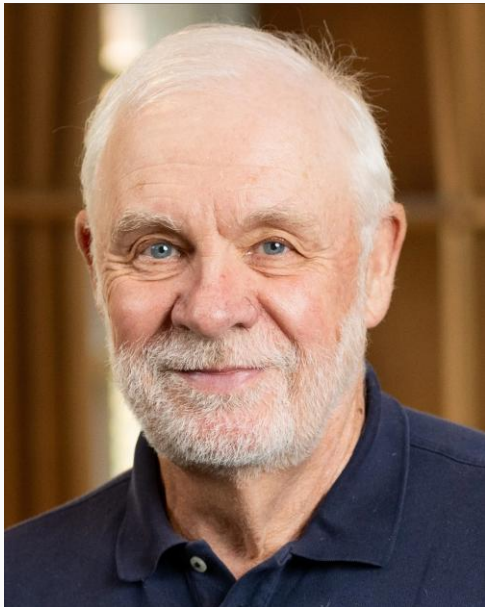


<https://src.isr.umich.edu/research/faculty-profiles/profiles/littleroderick-j/>

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Psychological Methods

<https://doi.org/10.1037/met0000695>

## Simulation Studies for Methodological Research in Psychology: A Standardized Template for Planning, Preregistration, and Reporting

Björn S. Siepe<sup>1</sup>, František Bartoš<sup>2</sup>, Tim P. Morris<sup>3</sup>, Anne-Laure Boulesteix<sup>4, 5</sup>,  
Daniel W. Heck<sup>1</sup>, and Samuel Pawel<sup>6, 7</sup>

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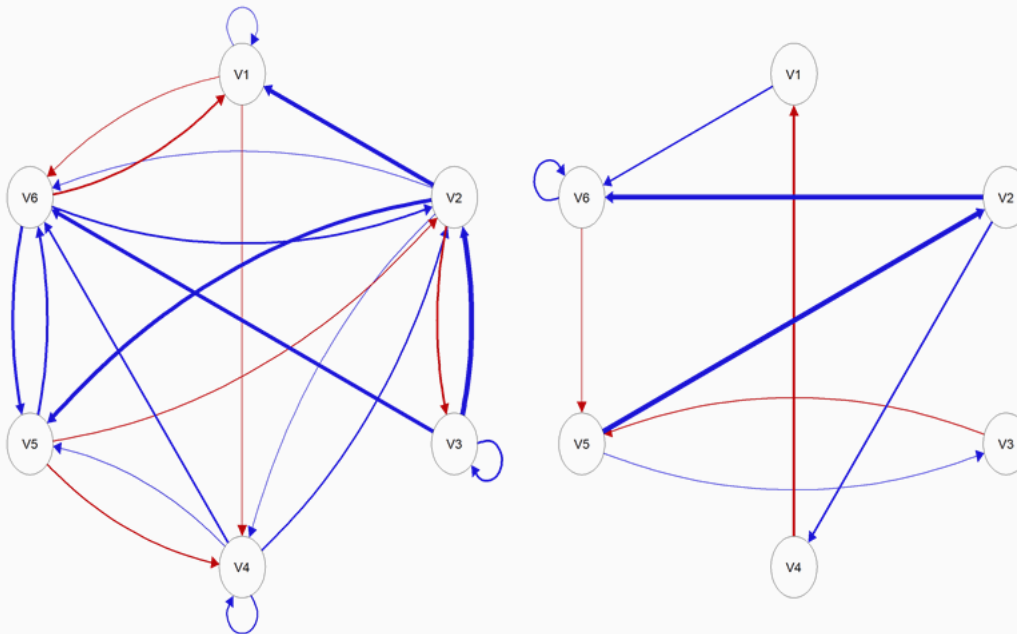
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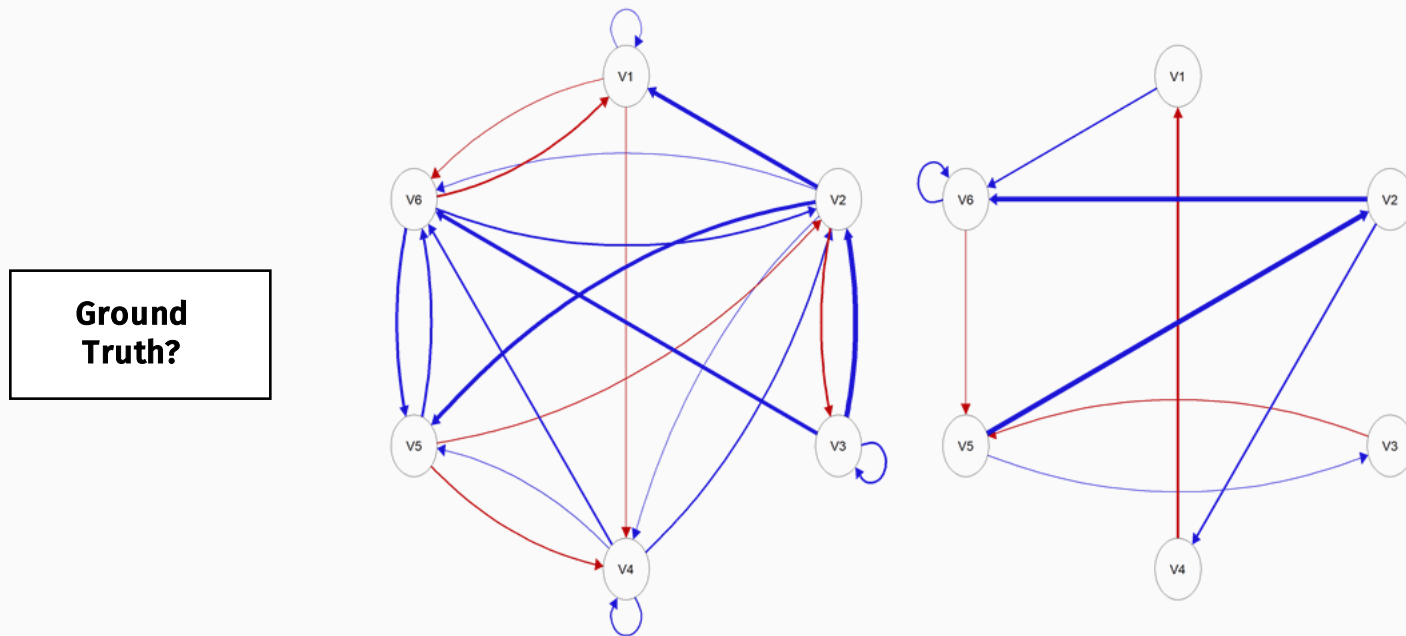
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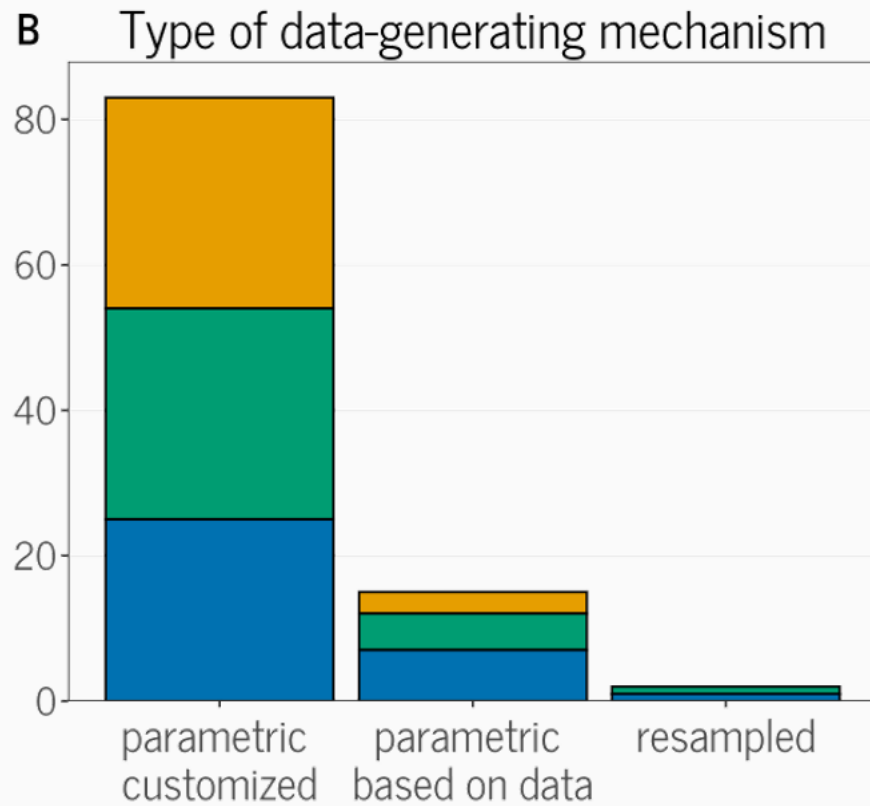
What is a useful/realistic data-generating mechanism?



➡ Hard to understand when performance differences actually matter

# The basis of our DGMs

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# What about benchmarking?

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## Computer Vision/ML



Source: Steni Sebastian, Medium

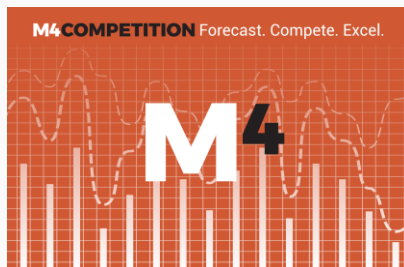
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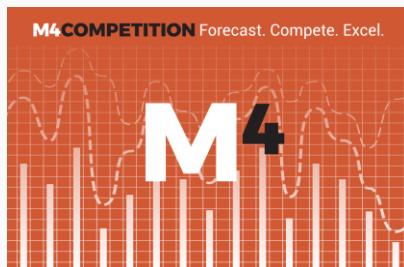
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### EDITORIAL

## Benchmarking comes of age

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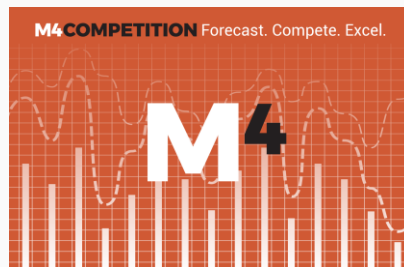
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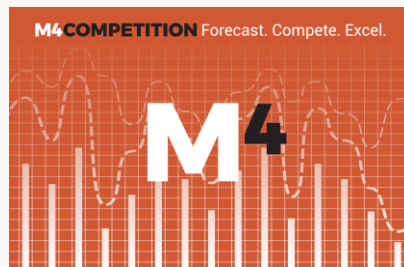
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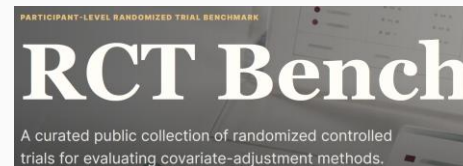
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Item Response Warehouse



# Introducing openESM





## Database

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- Harmonized
- Rich metadata



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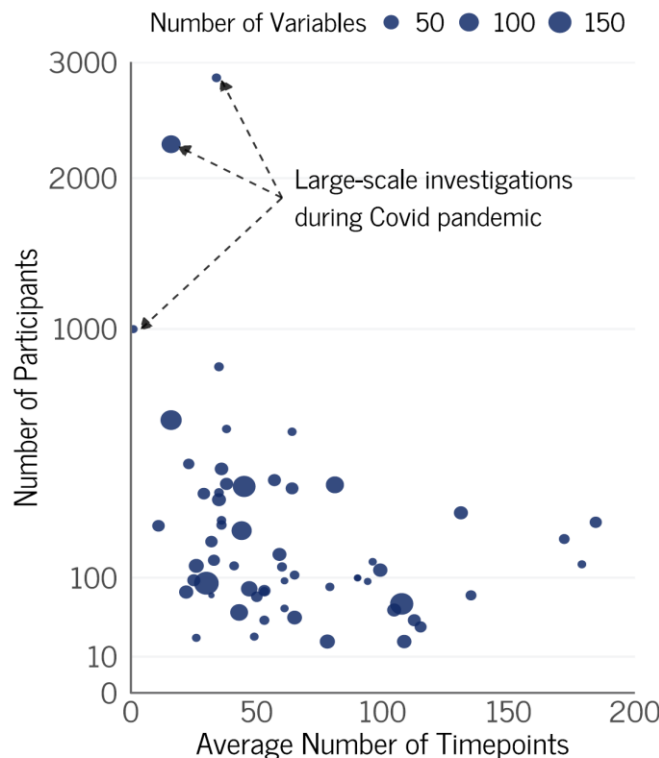
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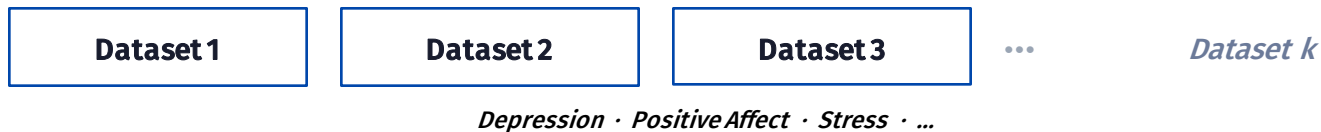
### A. Timepoints, Variables, and Participants



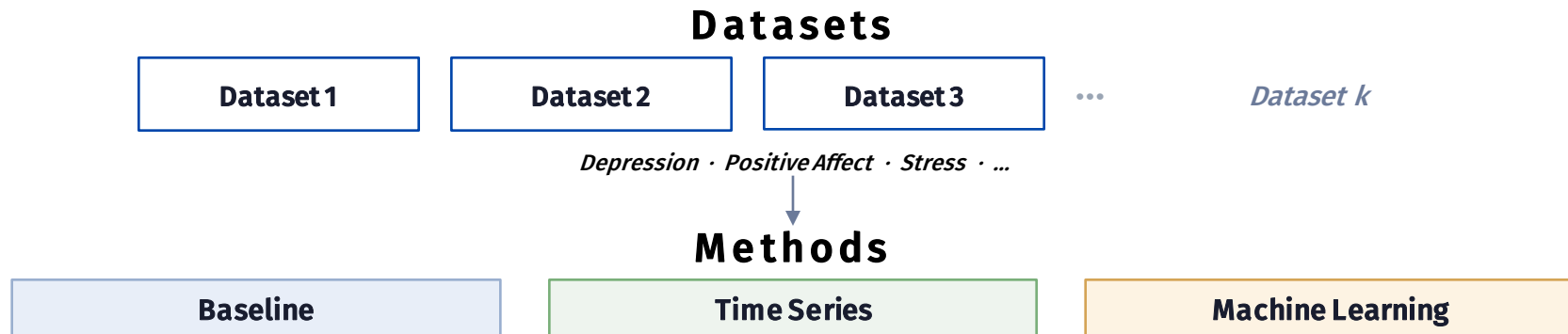
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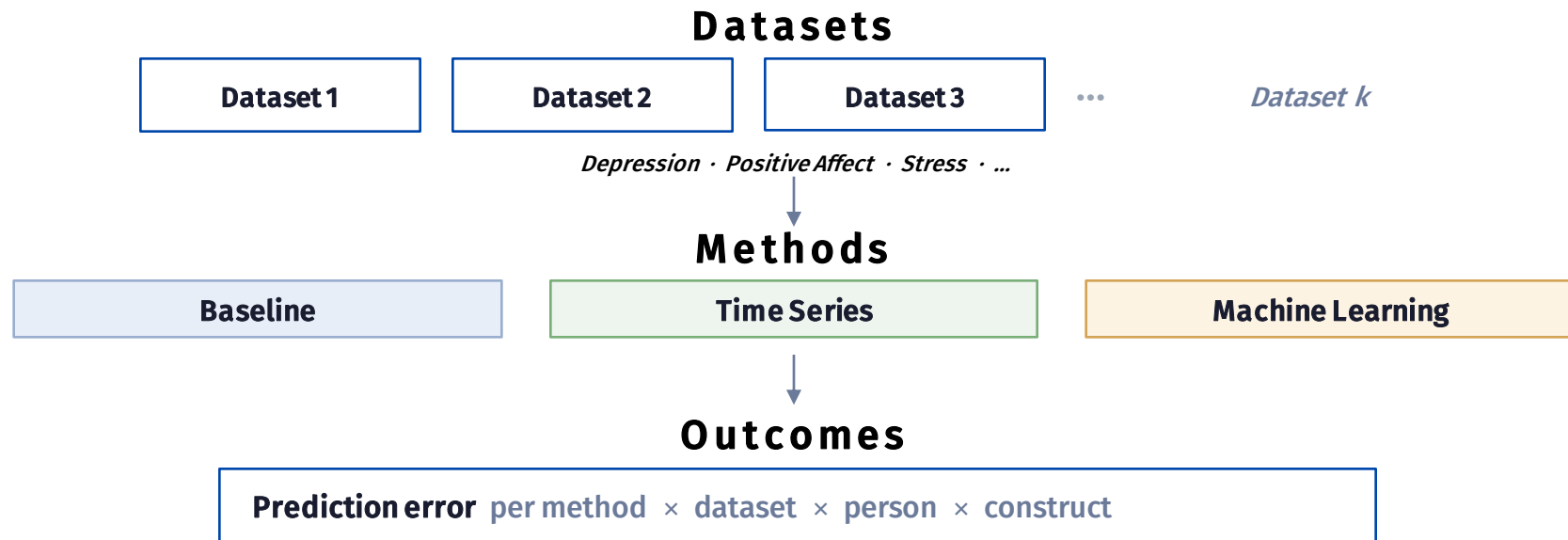
## Datasets



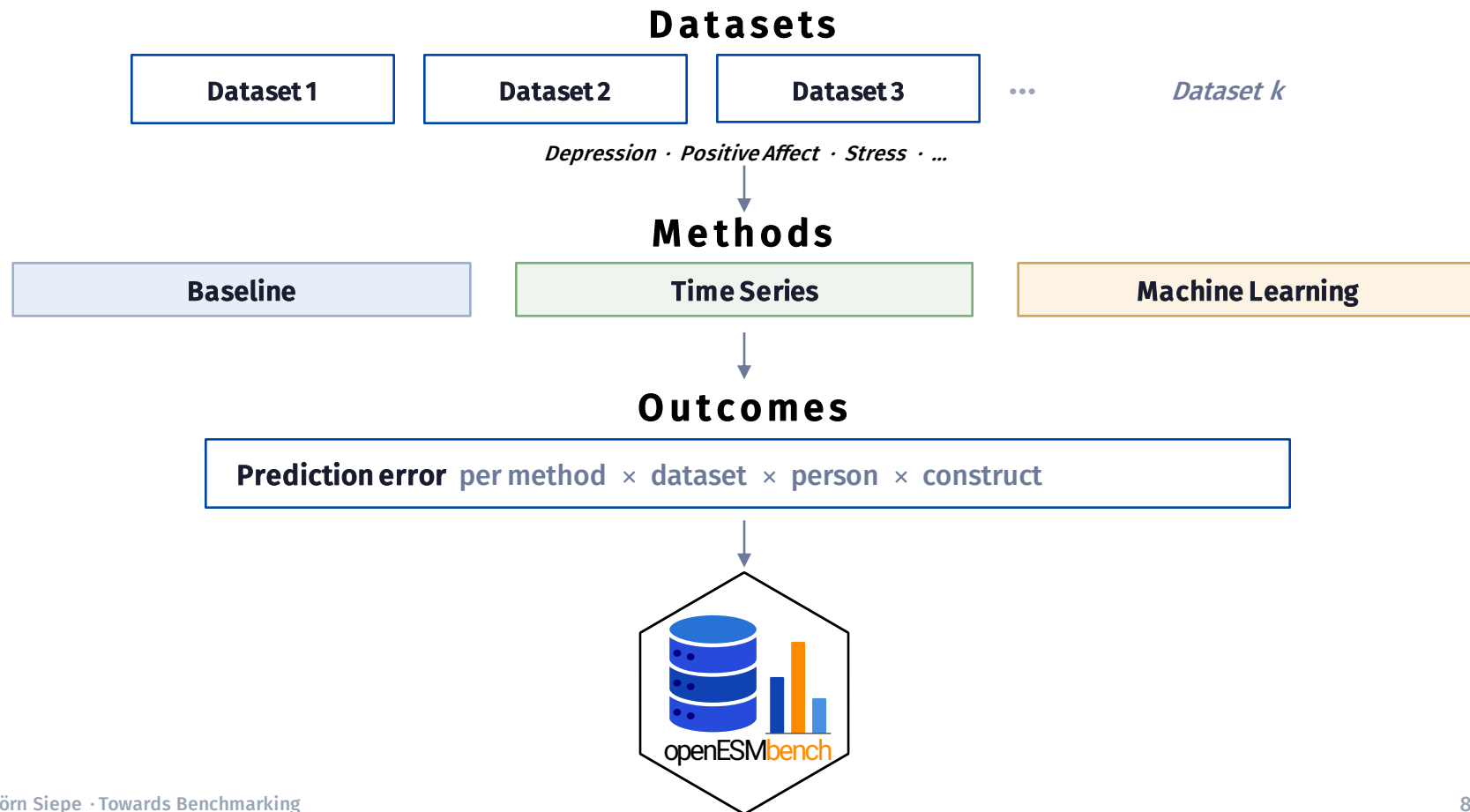
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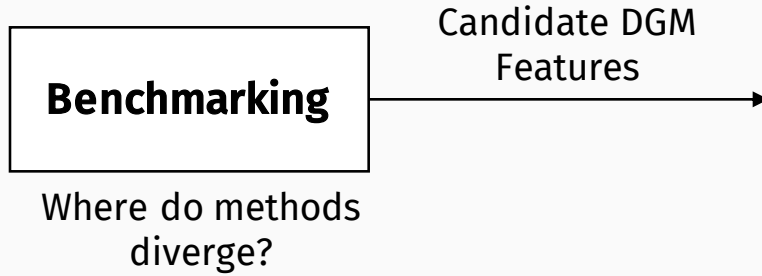


# Closing the loop: Benchmarking and simulations

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**Benchmarking**

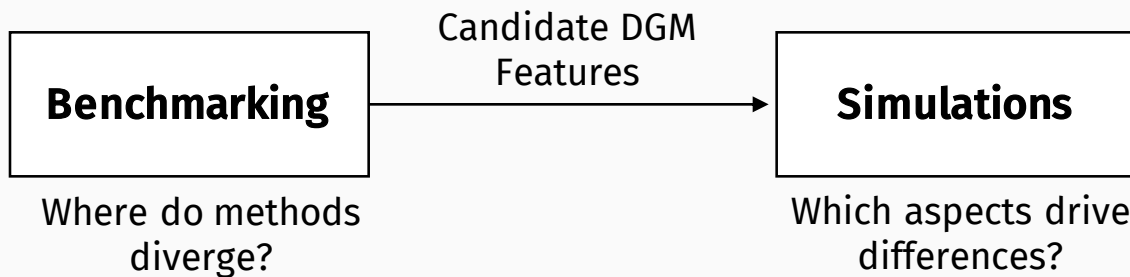
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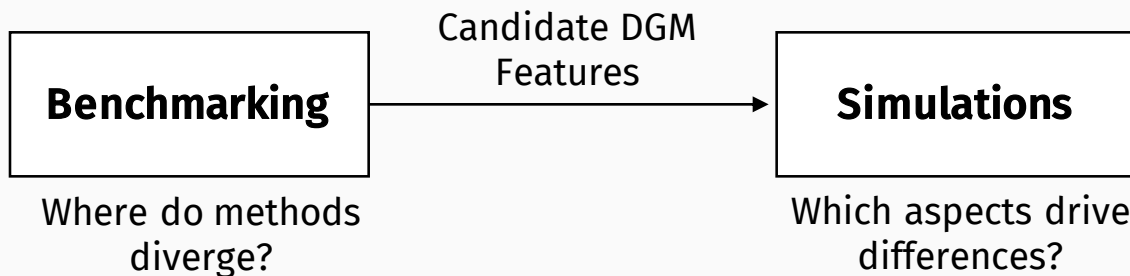
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**On the role of benchmarking data sets and simulations in method comparison studies**

[Sarah Friedrich](#) [✉](#) [Tim Friede](#)

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## Statistical Parametric Simulation Studies Based on Real Data

Christina Sauer, F. Julian D. Lange, Maria Thurow, Ina Dormuth, Anne-Laure Boulesteix [✉](#)

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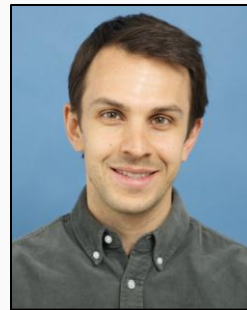
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## Takeaways

1. Psychology is **ready for benchmarking**
2. Let's **not limit ourselves** to one way of investigating methods

# Co-Authors



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# Thank You!

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<https://openesmdata.org>



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<https://bsiepe.github.io>



[bsiepe.bsky.social](https://bsiepe.bsky.social)

## Slides & Papers



# Resources

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